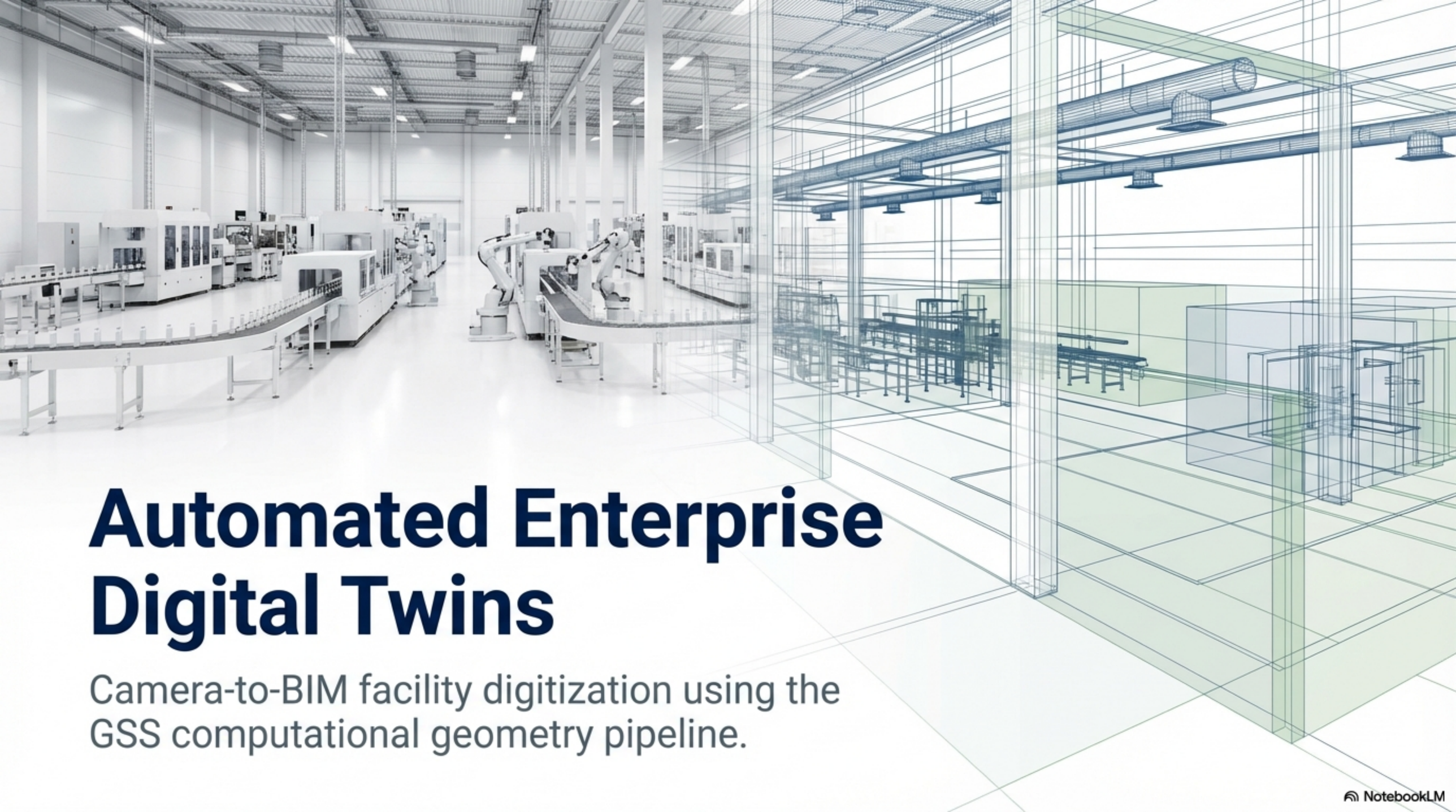


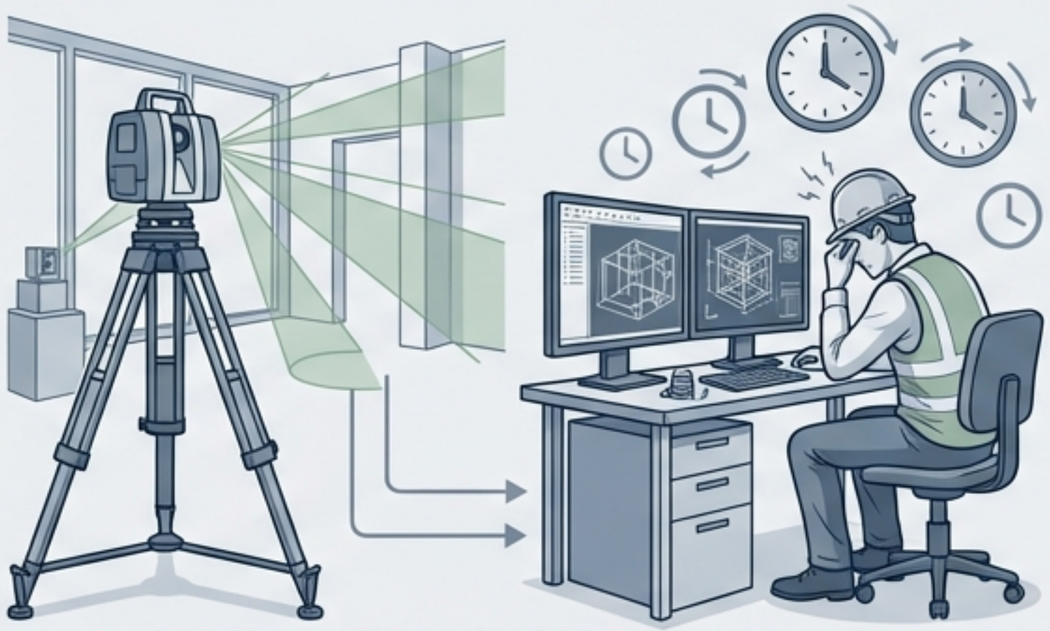
Automated Enterprise Digital Twins

Camera-to-BIM facility digitization using the GSS computational geometry pipeline.

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The Traditional Approach (Scan-to-BIM):

Requires expensive LiDAR scanning operations followed by weeks of manual 3D modeling.



Missing Scalable Solution

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Rapid facility digitization from standard video

The GSS (3D Gaussian Splatting to BIM) engine replaces manual modeling with an automated computational geometry pipeline. It transforms standard camera video into structured architectural data, eliminating the need for LiDAR.



Supported Pipeline Modes

1. Standard: Video → 3DGS → IFC BIM → Mesh

2. PlanarGS Integration: State-of-the-art AI co-planarity awareness

3. Exterior: Ground separation and facade extraction

4. Import Direct: Processing existing 3DGS point clouds

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Stage 1: Structuring spatial noise into architectural planes architectural planes

The pipeline ingests raw 3D Gaussian Splatting data—often millions of unstructured points—and applies robust Open3D RANSAC algorithms combined with alpha-shape polygonization.

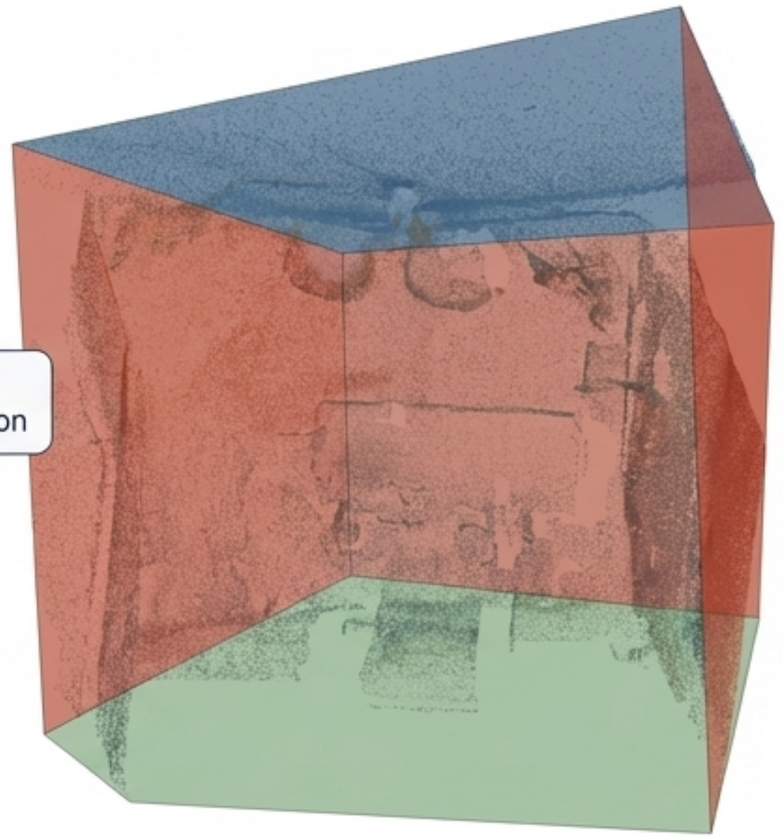
- **Input**: High-density spatial capture (e.g., 21.4M points).
- **Processing**: Automatic detection of planar boundaries.
- **Output**: Unstructured noise is grouped into discrete, geometric planes.

Before: Raw 3D Gaussian Splatting Data



RANSAC
Plane Detection

After: Structured Architectural Planes



Before: Raw 3D Gaussian Splatting Data

After: Structured Architectural Planes

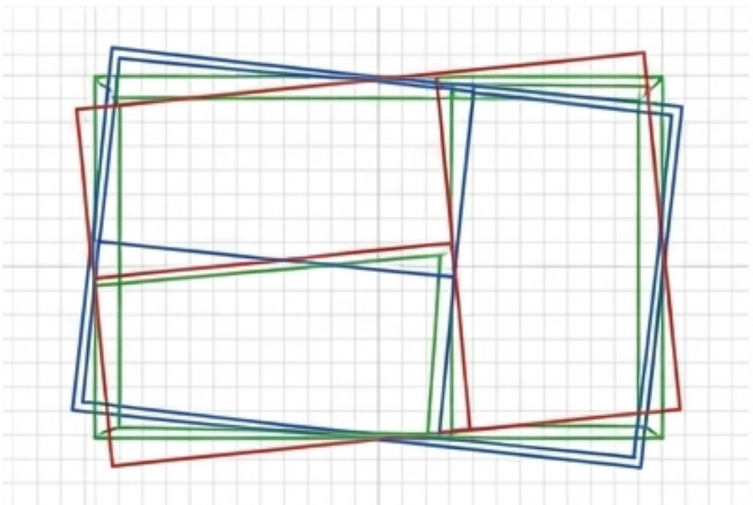
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Stage 2: Enforcing structural integrity via Manhattan alignment

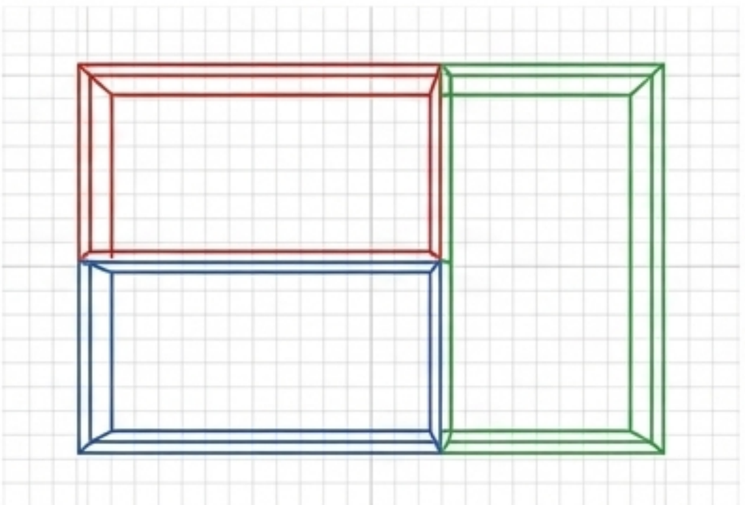
Extracted planes from real-world captures are rarely perfect. The GSS Regularization Engine applies strict architectural rules to generate pristine, enterprise-ready geometry.

- **Axis Alignment**: Snapping wall normals to precise Manhattan axes (Y-up coordinates).
- **Height Clustering**: Standardizing floor and ceiling elevations.
- **Intersection Trimming**: Automatically snapping wall endpoints to perfect 90-degree corners.
- **Wall Closure**: Synthesizing missing architecture from detected floor boundaries.

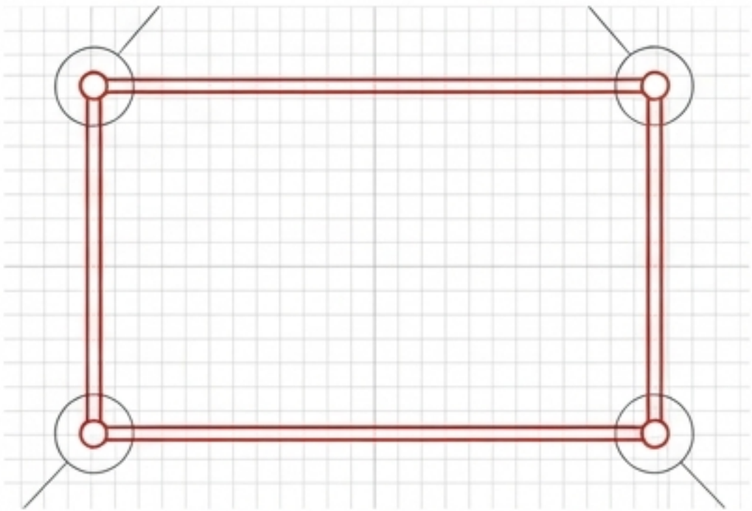
Raw RANSAC



Regularized



Manhattan Center-lines

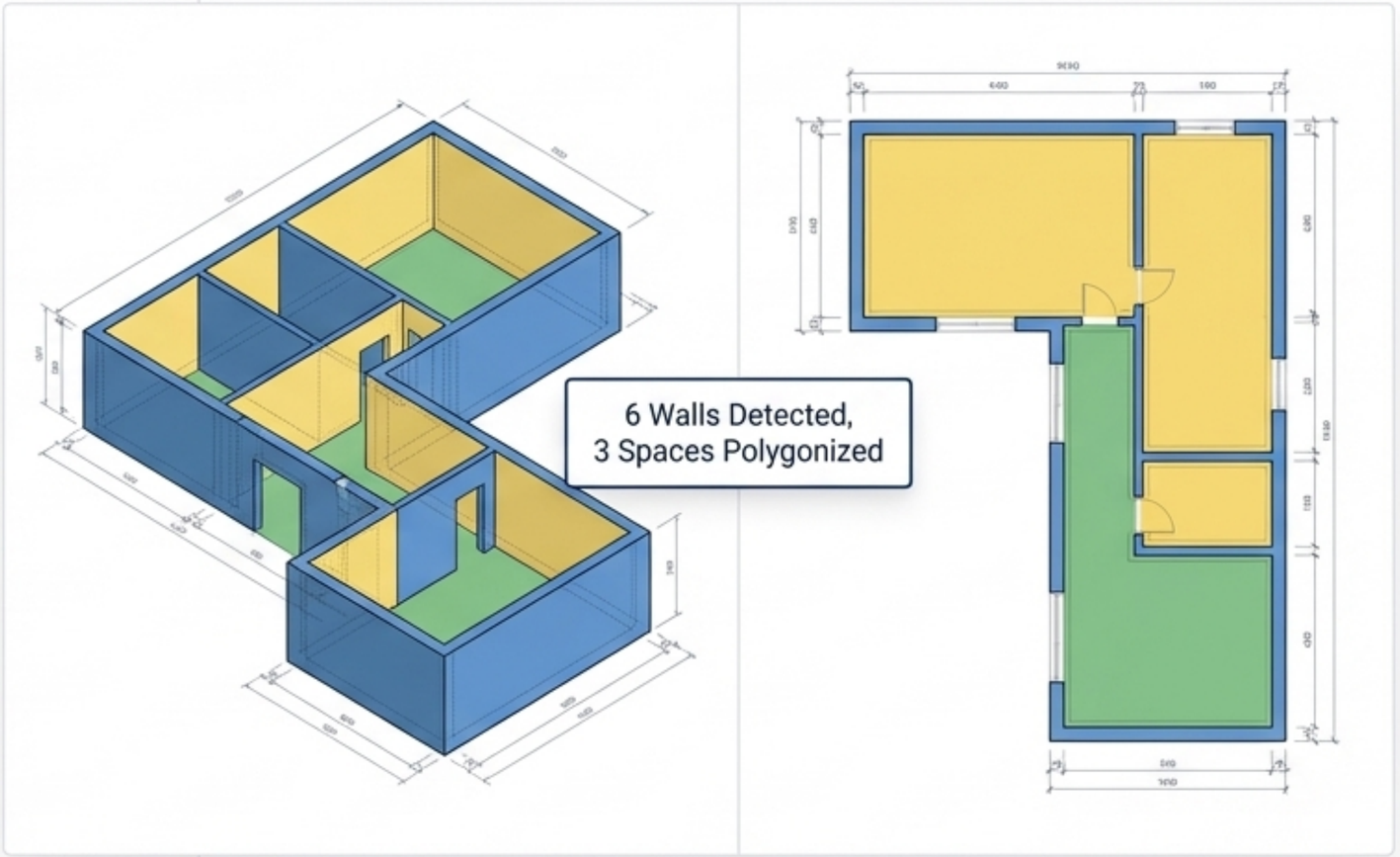


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Resolving complex facility footprints automatically

Enterprise facilities are rarely simple boxes. The pipeline utilizes a polyline wall builder and space polygonization to accurately reconstruct multi-room and non-standard architectures.

- Detects and segments multi-point polyline walls.
- Automatically polygonizes interior spaces for accurate room volume extraction.
- Successfully maps L-shaped and polygonal building structures without manual intervention.

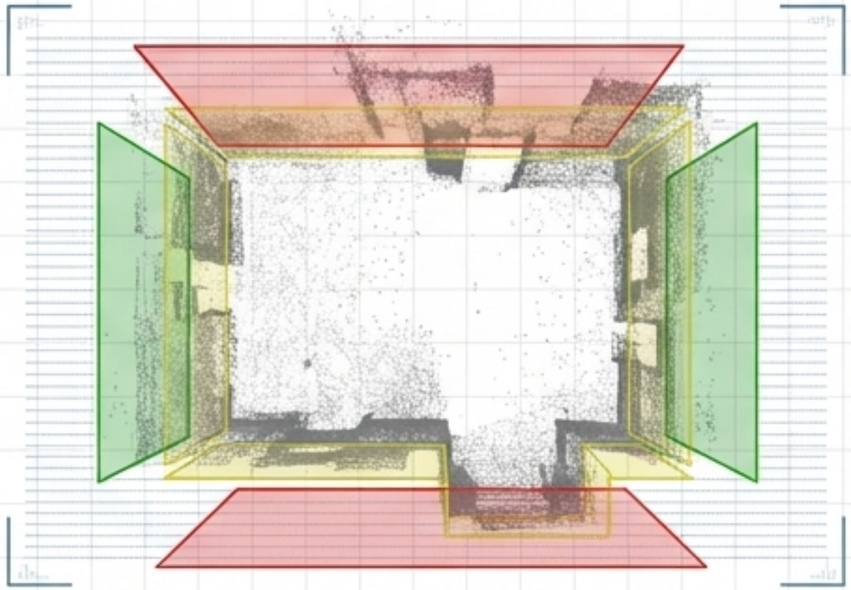


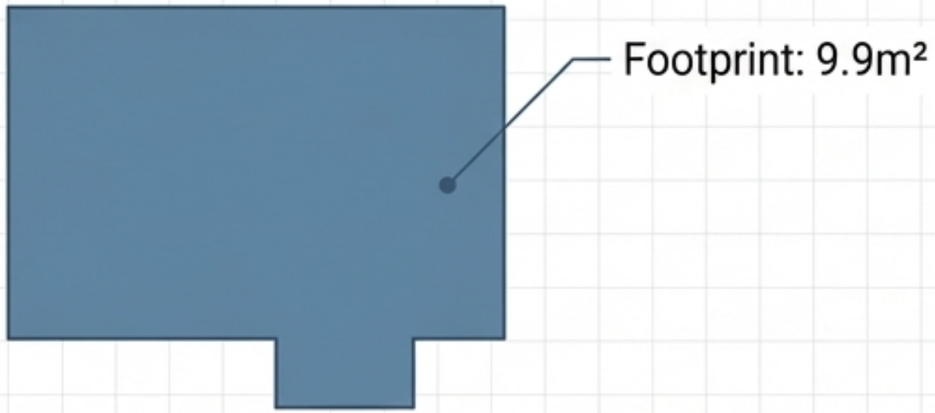
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Advanced architectural intelligence and feature extraction

The pipeline extracts deep semantic meaning from the spatial data, preparing the model for advanced facility management.

- **Opening Detection:** Integrates Cloud2BIM logic, analyzing point cloud depth discontinuities to automatically locate and measure doors and windows.
- **Scale Estimation:** Autonomously calculates absolute scale using floor-ceiling height deltas and wall Y-extents.
- **Exterior Structuring:** Separates ground planes, clusters normal-grouped vertical facades, and extracts exact 2D building footprints.

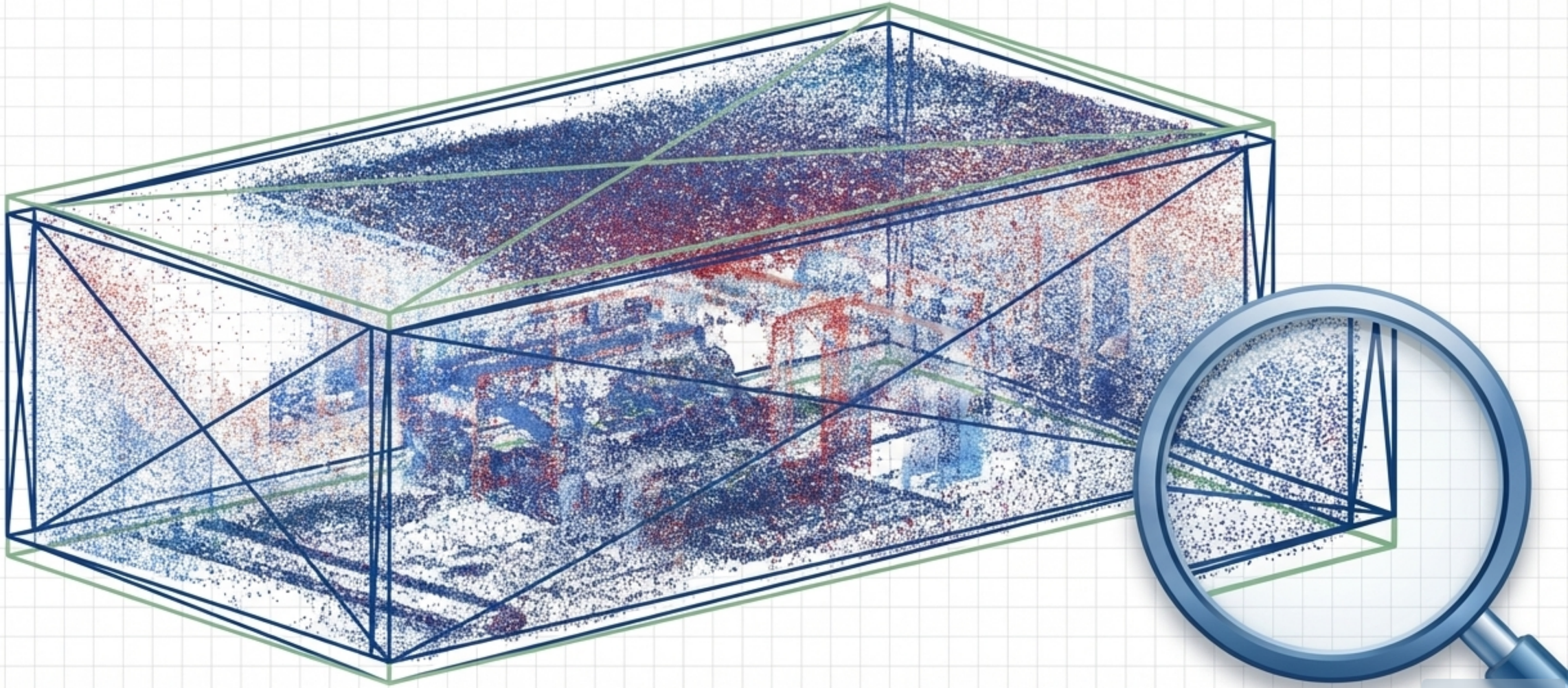




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Empirical proof of coordinate accuracy

Visual validation confirms exact spatial alignment between the raw 3DGS point cloud and the generated Industry Foundation Classes (IFC) geometry. The automated output perfectly encapsulates the physical capture.

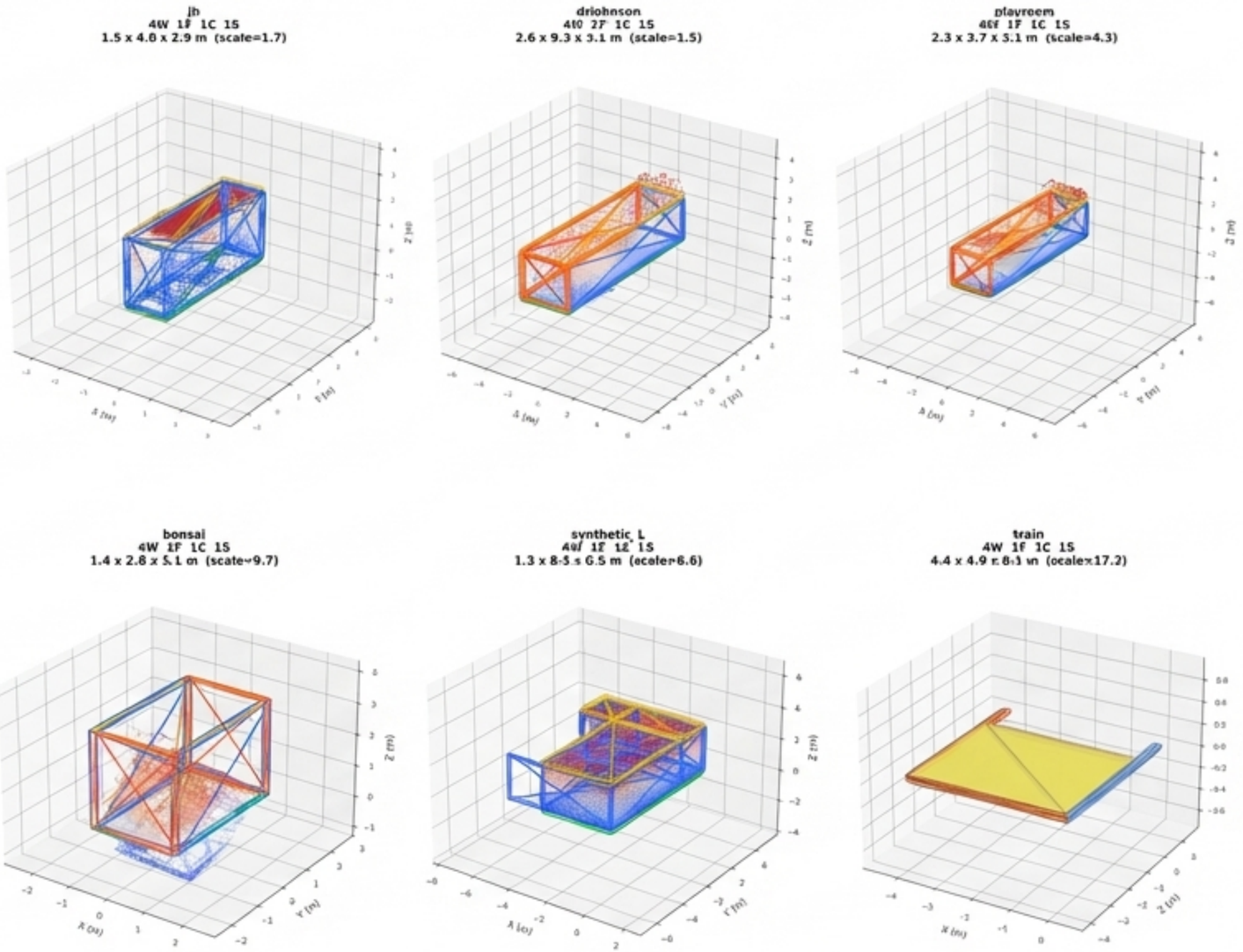


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Battle-tested across diverse physical environments

The GSS pipeline is an enterprise-grade engine, proven across 9 diverse structural datasets ranging from simple rooms to complex, multi-scale environments.

- **Robust Architecture**: Supported by 365 passing unit tests validating everything from geometry merging to precise IFC data structuring.
- **Consistent Output**: Generates accurate 4-wall, 1-floor, 1-ceiling (4W 1F 1C) baselines consistently across diverse captures.

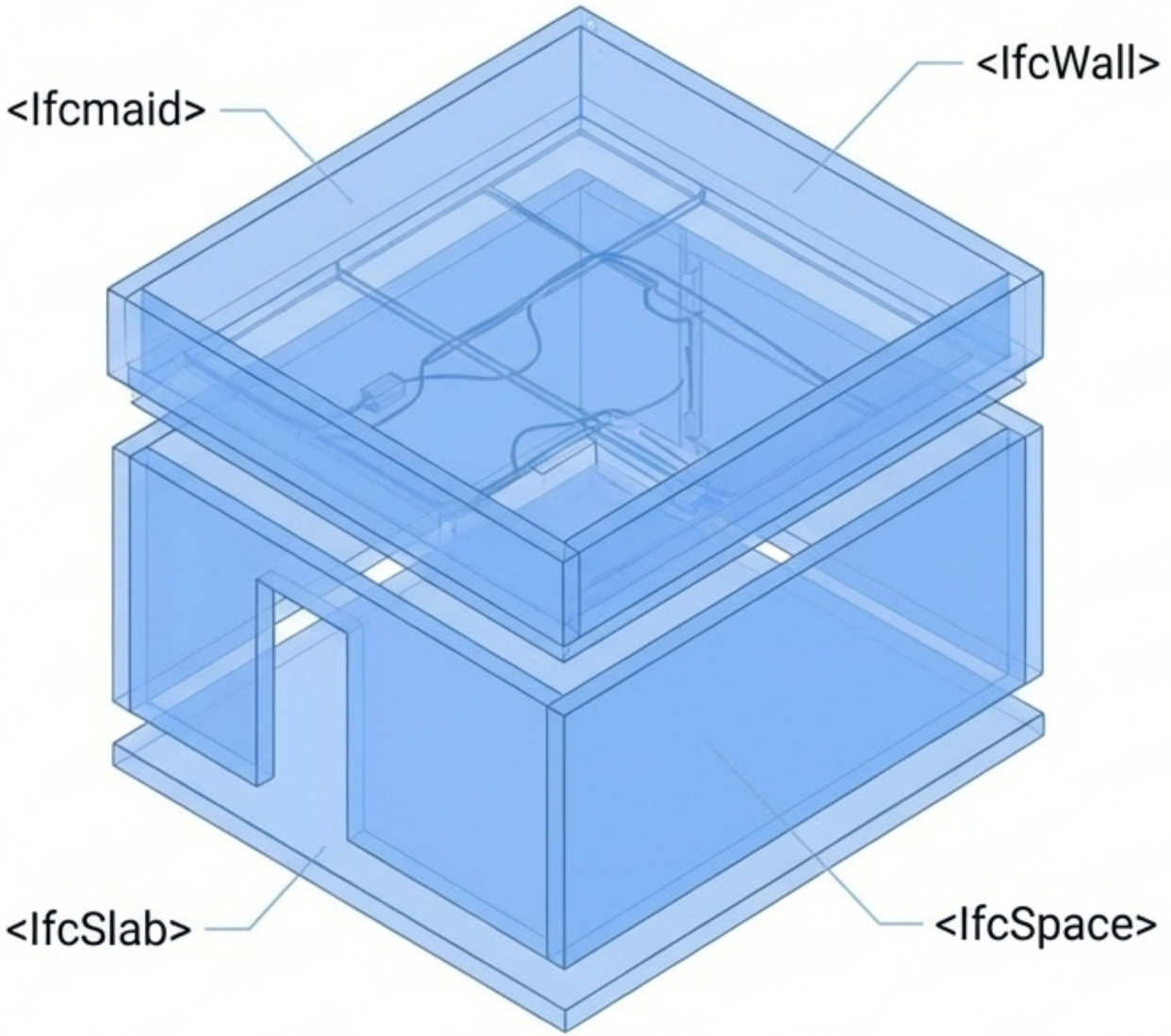


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Generating native industry-standard BIM

The final output is not just a visual mesh, but a fully structured Industry Foundation Classes (IFC4) file. This acts as the foundational layer for P&G's facility management systems.

- Generates native IfcWall, IfcSlab, IfcSpace, and IfcOpeningElement entities.
- Preserves exact architectural hierarchy and metadata.
- Provides a seamless bridge between visual reality capture and engineering-grade CAD systems.

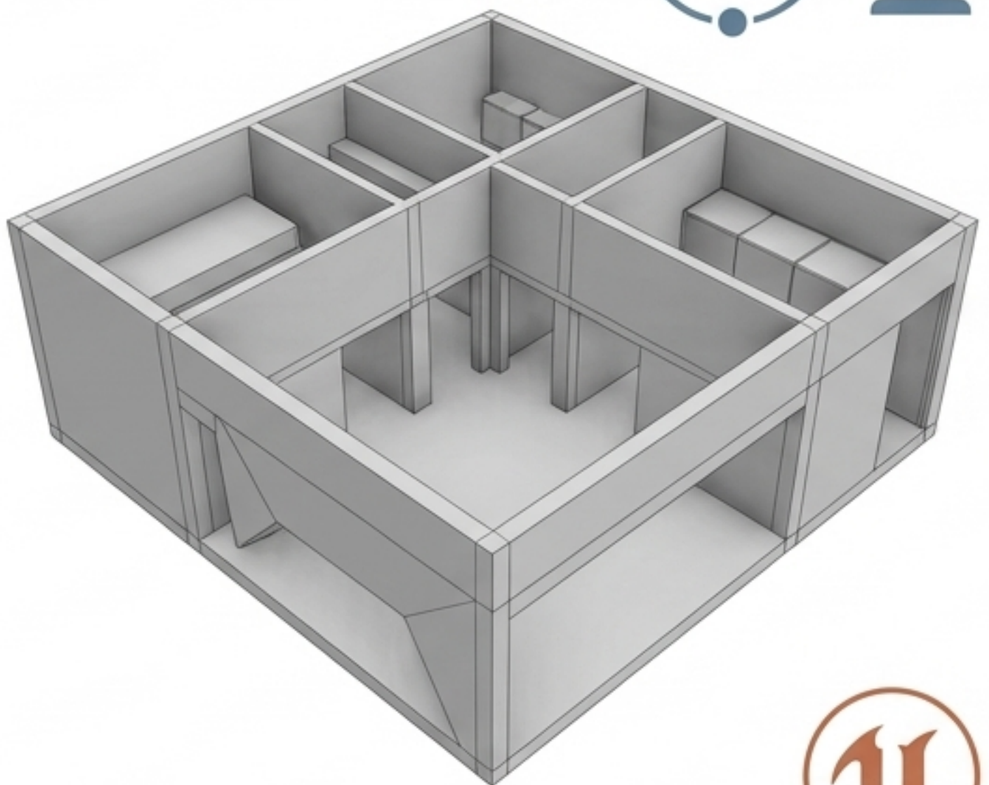


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Ready for industrial simulation and spatial computing

The generated IFC data is automatically transcoded into highly optimized, cross-platform 3D formats, ready for immediate deployment in advanced enterprise workflows.

- **USD / USDC:** Y-up coordinate formatted natively for NVIDIA Omniverse and Isaac Sim. Essential for automated robotics pathfinding and supply chain simulation.
- **USDZ:** Packaged with POSIX paths for immediate AR deployment on Apple Vision Pro.
- **GLB:** Z-up to Y-up sanitized meshes for instant viewing in web dashboards and Unreal Engine.



USD / USDC

USDZ

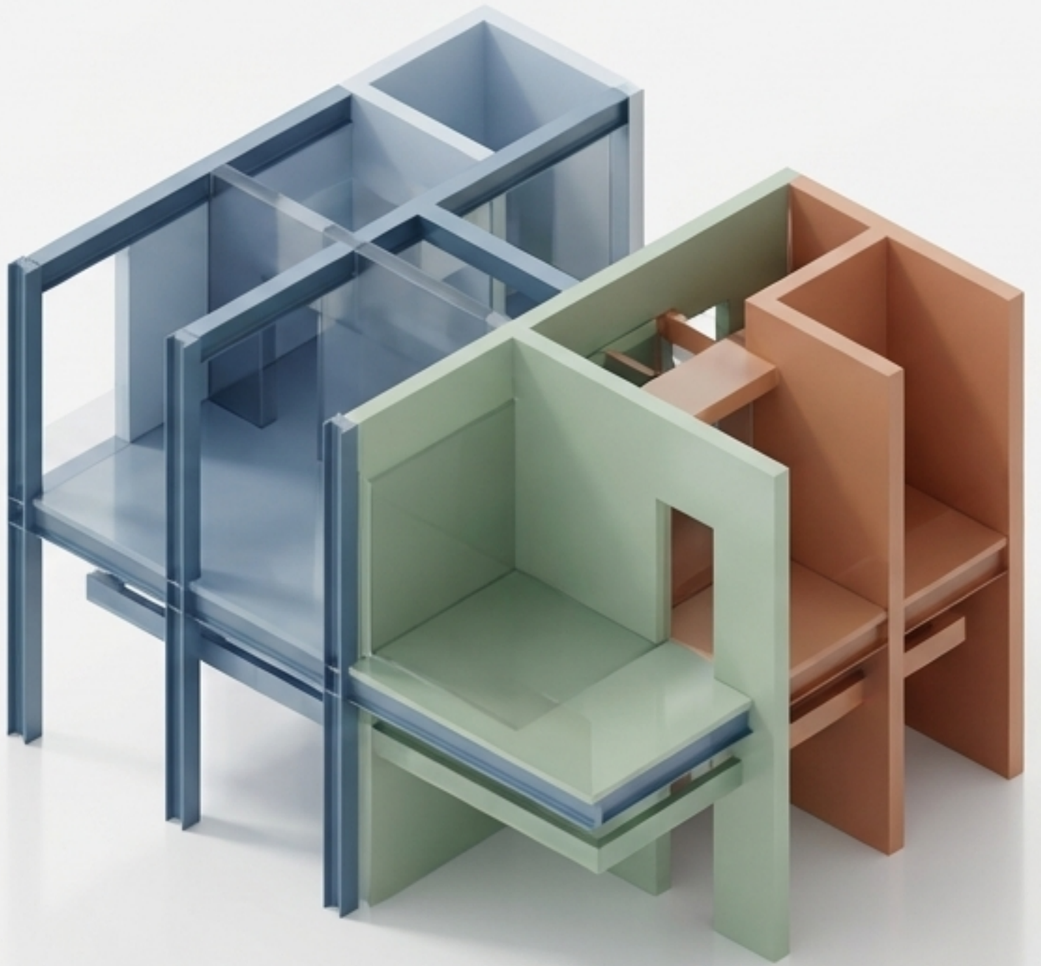
GLB

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Accelerating operations through automated digitization

By replacing weeks of manual LiDAR processing with the automated GSS camera-to-BIM pipeline, P&G can rapidly scale its digital twin infrastructure.

Accurate, simulation-ready spatial assets can now be generated on-demand, powering the next generation of manufacturing efficiency, robotic automation, and facility management.



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